Saransh Jajoo

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IoT | Embedded Systems | AI

SUMMARY

Computer Science Engineering undergraduate with interests in Embedded Systems, IoT, Robotics, Artificial Intelligence. Passionate about designing and developing innovative engineering solutions that combine hardware and software to address real-world challenges while continuously learning and applying emerging technologies.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, SQL, Java
Embedded Systems & IoT: ESP32, ESP8266, Raspberry Pi, Embedded C/C++, GPIO, I2C, PWM, ADC, Sensor Interfacing, IoT System Development
Hardware & Electronics: Microcontrollers, Circuit Design, Actuators, Embedded Debugging, Hardware-Software Integration
Frameworks & Libraries: NumPy, Pandas, Scikit-Learn, Streamlit, FastAPI
Developer Tools: Git, GitHub, Linux, Ubuntu, Arduino IDE, PlatformIO, VS Code, Jupyter Notebook, Canva, Blynk, ThingSpeak

EXPERIENCE

Embedded Systems Intern — iThink Software Labs Jun 2025 – Sept 2025
- Developed industrial P10 RGB LED display systems using ESP8266 and ESP32 microcontrollers.
- Improved display readability by 30–40% through custom pixel-level rendering optimization.
- Integrated real-time web data feeds with sub-200 ms refresh latency.

Member — e-Yantra Robotics Club, ADCET 2024 – Present
- Designed, built, and programmed autonomous mobile robots using embedded systems and sensor-based control.
- Developed line-following, obstacle avoidance, and autonomous navigation robots using microcontrollers and real-time sensor feedback.
- Implemented control algorithms for precise robot navigation, path tracking, and motor control.

PROJECTS

PunarJAL — IoT-Enabled Greywater Management System
- Developed an IoT-enabled greywater treatment and monitoring solution focused on sustainable water reuse.
- Implemented automated sensing and treatment control supporting **24x7 water quality monitoring** through real-time environmental sensors.
- Designed a scalable architecture suitable for residential, institutional, and community-scale deployment.

SensiCold — Smart Portable IoT-Based Cold Storage System
- Architected a modular portable cold-storage solution with a **crop-specific, one-unit-per-crop design** enabling independent environmental control and scalable deployment.
- Built an **ESP32-powered embedded control system** interfacing **4+ environmental sensors** for real-time monitoring, actuator control, and adaptive storage condition management.
- Developed a cross-platform IoT monitoring application with **crop configuration, and multi-unit supervision**, alongside a virtual engineering prototype for hardware validation.

AURA-CX — AI-Powered Omnichannel Customer Experience Orchestrator
- Architected an AI-powered customer support platform integrating **4+ communication channels** into a unified support command center.
- Developed vector-similarity-based identity resolution capable of creating **360-degree customer profiles** across fragmented communication channels.
- Engineered sentiment analytics and clustering workflows targeting **almost 180 second response times**, **70% Tier-1 support automation**, and **15% churn reduction**.

ACHIEVEMENTS

• Samsung SFT 2025 Top 100 • PuneAgri Hackathon 2025 Finalist • SIH 2025 Grand Finalist • Anveshan 2026 National Finalist • ADCET Hackathon 2025 Winner • Idea 2.0 Top 100

EDUCATION

Annasaheb Dange College of Engineering and Technology (ADCET), Ashta 2023 – Present
B.Tech in Computer Science Engineering CGPA: **8.35/10**